

Choronos-Grid: Temporal Energy Transport, Search, and CSP Generation

CS 188 AI Project (Milestone 1 & 2)

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Project Overview & Core Architecture

The Envoirement:

An automated reactor facility represented as an $N \times N$ discrete grid operating across a repeating temporal cycle $t \in \{0, 1, \dots, T-1\}$.

Core Challenges:

- **Dynamic Temporal Topology:** Transitions depend on coordinates (x,y) and cyclic time $t \pmod T$.
- **Resource Budgets:** A non-renewable battery reservoir ($E_{\max}$) is depleted by movement, wait actions, and ion storms, while single-use solar pads replenish energy.
- **Non-Standard Constraints:** Resists canned template matching by coupling spatial paths with temporal hazard schedules and resource invariants.

Milestone 1 - Reactor Floor Generator

- **CSP Formulation:**

- **Variables:** Grid cells $V_{\{x,y\}}$ for $x, y \in \{0, 1, \dots, N-1\}$.
- **Domains:** $D(V_{\{x,y\}}) = \{\text{Empty}, \text{Wall}, \text{IonStorm}, \text{SolarRecharge}\}$.

- **Structural Constraints:**

- **Terminal Safety:** Cells $(0,0)$ and $(N-1, N-1)$ are permanently assigned as Empty.
- **Quotas:** Exactly $\lfloor 0.20N^2 \rfloor$ cells are designated as Wall, and exactly 3 cells are designated as SolarRecharge.
- **Anti-Adjacency:** No SolarRecharge cell may share an orthogonal or diagonal edge with an IonStorm cell.

Milestone 1 - Forward Checking & Heuristics

- **Direct Forward Checking:** Bypasses complex AC-3 propagation; prunes immediate unassigned neighbors' domains directly upon variable assignment.
- **Heuristic Strategies:**
 - **MRV (Minimum Remaining Values):** Selects the unassigned cell with the smallest remaining domain size.
 - **Degree Heuristic:** Breaks ties by choosing the variable with the highest number of unassigned neighbors.
- **Global Reachability Validation:** Concludes backtracking with an unweighted state-space BFS to ensure a valid energy-feasible path exists from start to finish.

Resolving Temporal Deadlocks (Project Addendum)

- **Addressing Coordinate Collapse:** Under initial equations, boundary nodes synchronized to block simultaneously at $t \equiv 0 \pmod{3}$. Applied official fixes:
 - **Static Safety Zones:** Terminal cells $(0,0)$ and $(N-1, N-1)$ are completely exempt from dynamic hazards ($\text{Blocked}(0,0,t) = \text{False}$).
 - **1-Indexed Offset Equation:** Hazard scheduling updated to $\text{Blocked}(x,y,t) \Leftrightarrow ((x+1)(y+1)+t) \pmod{3} == 0$ to stagger phase alignments.
 - **Constraint (e) Clarification:** Permits a Wait action while surrounded temporarily, provided at least one exit opens at step $t+1$.

Milestone 2 – Multi-Dimensional Pathfinder State Space

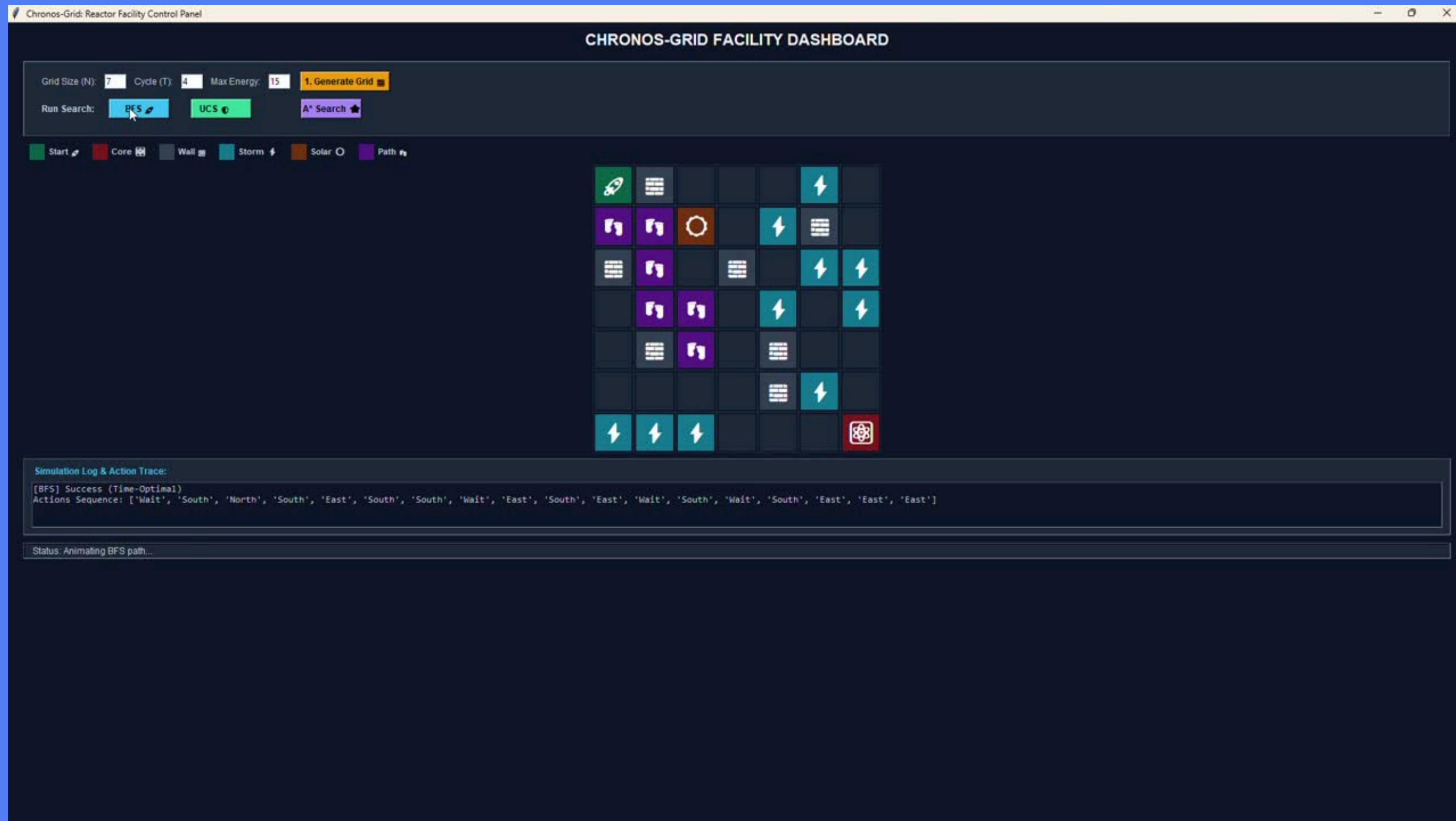
- **Augmented State Tuple:**
 - $s = (x, y, \tau, e, \text{solar_mask})$
 - $\tau = t(\text{mod } T)$ (cyclic temporal phase).
 - $e \in \{0, \dots, E_{\text{max}}\}$ (battery reserves).
 - **solar_mask (3-bit integer tracking single-use recharge pad consumption).**
- **Action Space ($\mathcal{A}$):** {North, South, East, West, Wait}, each primitive action advancing the global clock $t \rightarrow (t+1) (\text{mod } T)$.

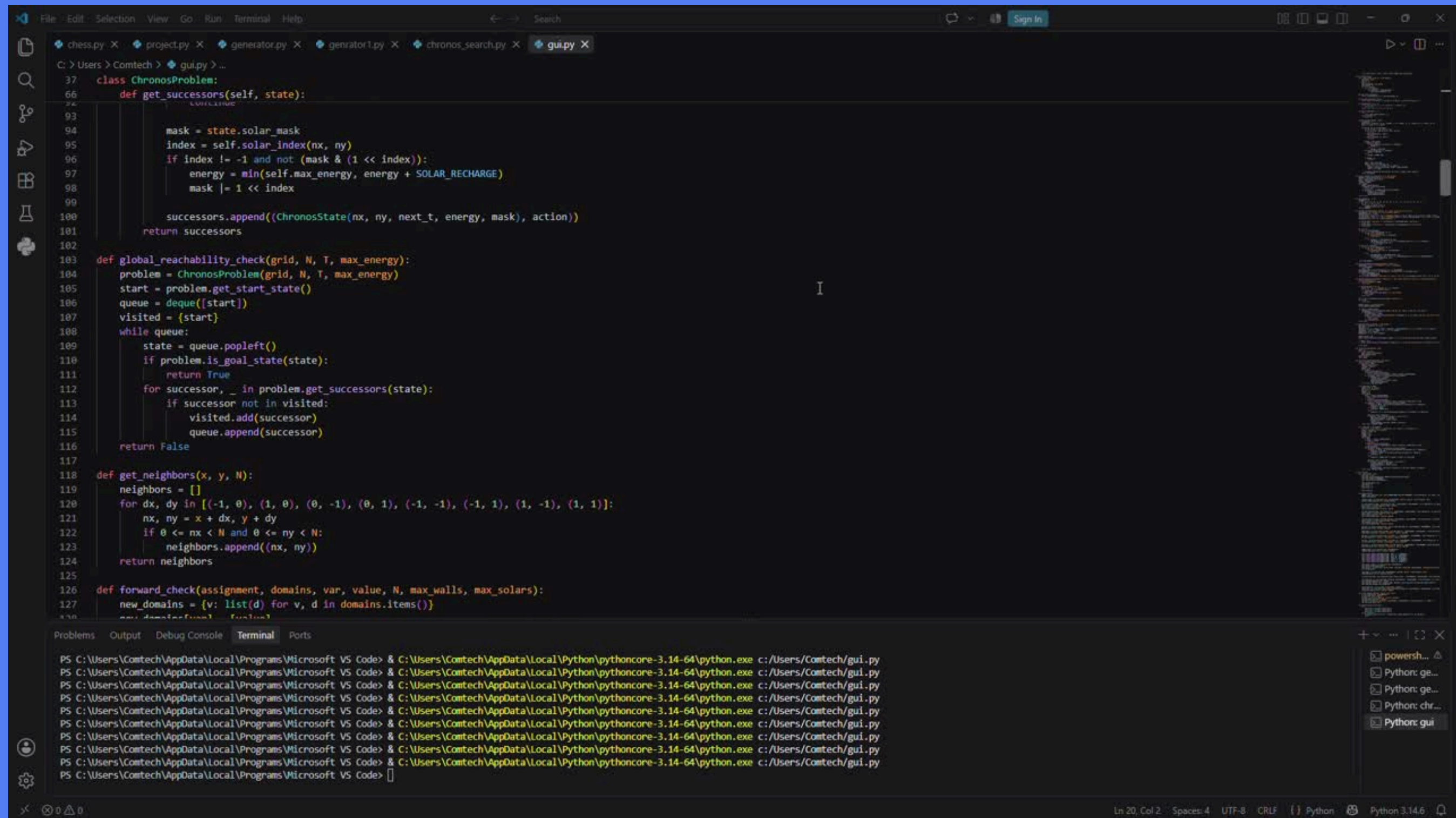
Milestone 2 – Search Algorithms Implemented

- **Breadth-First Search (BFS):**
 - Finds the minimal time-step trajectory while ignoring energy limits.
- **Uniform Cost Search (UCS):**
 - Finds the minimal total energy expenditure trajectory using priority queues and step-cost tracking.
- **A* Graph Search:**
 - Evaluated using consistent heuristic h_1 (Manhattan distance divided by maximum velocity) and supported by an extensible closed-set framework with node-reopening options (`allow_reopen=True`).

Conclusion & Summary

- **Integrated Pipeline:** Successfully links automated CSP grid generation with multi-dimensional pathfinding algorithms.
- **Robust Verification:** Ensures structural quotas, anti-adjacency rules, hazard management, and energy preservation are fully satisfied.
- **Ready for Live Demo / Questions!**







Thank You